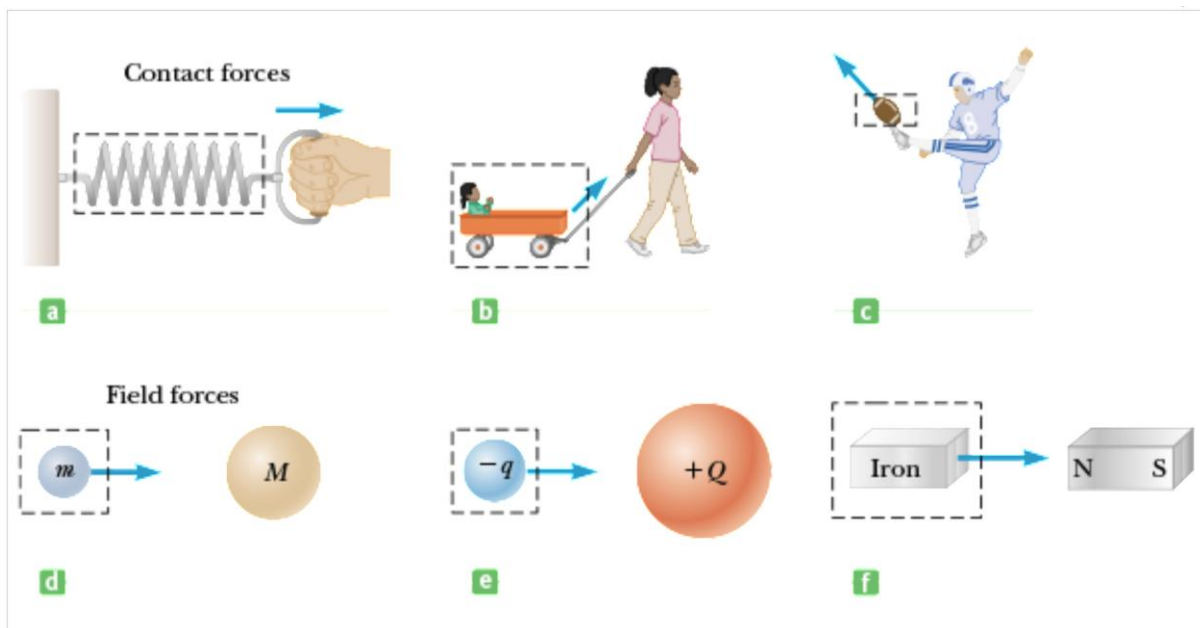


TUTORIAL CHAPTER 4: NEWTON'S LAWS OF MOTION

The three levels of problems are graded according to their difficulty. Straightforward problems are numbered in **black**, intermediate level problems are numbered in **blue**, and the most challenging problems are numbered in **red**.

4.3 Newton's Third Law

4. **Q/C** One or more external forces are exerted on each object enclosed in a dashed box shown in [Figure 4.2](#). Identify the reaction to each of these forces.



6. A freight train has a mass of $1.5 \times 10^7 \text{ kg}$. If the locomotive can exert a constant pull of $7.5 \times 10^5 \text{ N}$, how long does it take to increase the speed of the train from rest to 80 km/h ?

Ans.: 7.4 min

10. **V** A 5.0-g bullet leaves the muzzle of a rifle with a speed of 320 m/s . What force (assumed constant) is exerted on the bullet while it is traveling down the 0.82-m -long barrel of the rifle?

Ans.: $3.1 \times 10^2 \text{ N}$

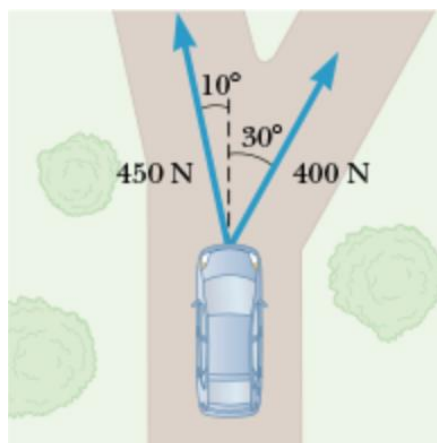
11. A boat moves through the water with two forces acting on it. One is a 2.00×10^3 -N forward push by the water on the propeller, and the other is a 1.80×10^3 -N resistive force due to the water around the bow.

- a. What is the acceleration of the 1.00×10^3 -kg boat?
- b. If it starts from rest, how far will the boat move in 10.0 s ?
- c. What will its velocity be at the end of that time?

Ans.: a. 0.200 m/s^2 ; b. 10.0 m ; c. 2.00 m/s

12. Two forces are applied to a car in an effort to move it, as shown in [Figure P4.12](#).

- a. What is the resultant vector of these two forces?
- b. If the car has a mass of 3 000 kg , what acceleration does it have? Ignore friction.



Ans.: a. 798 N at 8.79° to the right of the forward direction ; b. 0.266 m/s^2

15. After falling from rest from a height of 30.0 m , a 0.500-kg ball rebounds upward, reaching a height of 20.0 m . If the contact between ball and ground lasted 2.00 ms , what average force was exerted on the ball?

Ans.: $1.11 \times 10^4 \text{ N}$

18. What would be the acceleration of gravity at the surface of a world with twice Earth's mass and twice its radius?

Ans.: 4.90 m/s^2

4.5 Normal Forces

19. Calculate the magnitude of the normal force on a 15.0-kg block in the following circumstances:

- a. The block is resting on a level surface.
- b. The block is resting on a surface tilted up at a 30.0° angle with respect to the horizontal.
- c. The block is resting on the floor of an elevator that is accelerating upward at 3.00 m/s^2 .
- d. The block is on a level surface and a force of 125 N is exerted on it at an angle of 30.0° above the horizontal.

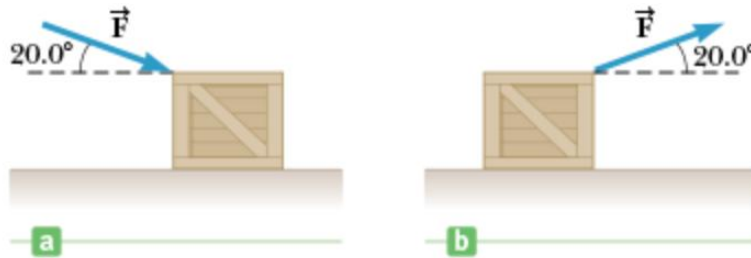
Ans.: a. 147 N ; b. 127 N ; c. 192 N ; d. 84.5 N

22. A student of mass 60.0 kg , starting at rest, slides down a slide 20.0 m long, tilted at an angle of 30.0° with respect to the horizontal. If the coefficient of kinetic friction between the student and the slide is 0.120 , find

- a. the force of kinetic friction,
- b. the acceleration, and
- c. the speed she is traveling when she reaches the bottom of the slide.

Ans.: a. 61.1 N ; b. 3.88 m/s^2 ; c. 12.5 m/s

23. A 1.00×10^3 -N crate is being pushed across a level floor at a constant speed by a force $\vec{F}$ of 3.00×10^2 N at an angle of 20.0° below the horizontal, as shown in [Figure P4.23a](#).

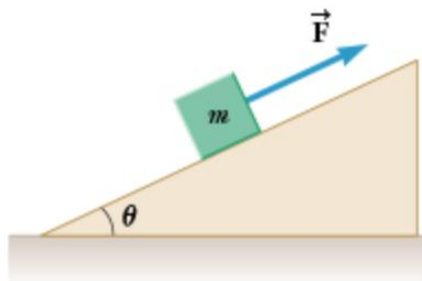


- a. What is the coefficient of kinetic friction between the crate and the floor?
- b. If the 3.00×10^2 -N force is instead pulling the block at an angle of 20.0° above the horizontal, as shown in [Figure P4.23b](#), what will be the acceleration of the crate? Assume that the coefficient of friction is the same as that found in part (a).

Ans.: a. 0.256 ; b. 0.510 m/s^2

24. A block of mass $m = 5.8 \text{ kg}$ is pulled up a $\theta = 25^\circ$ incline as in [Figure P4.24](#) with a force of magnitude $F = 32 \text{ N}$.

- a. Find the acceleration of the block if the incline is frictionless.
- b. Find the acceleration of the block if the coefficient of kinetic friction between the block and incline is 0.10 .



Ans.: a. 1.4 m/s^2 ; b. 0.49 m/s^2

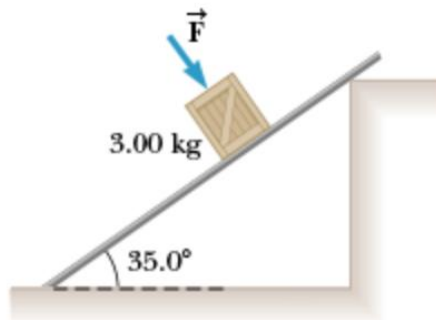
28. A block of mass 55.0 kg rests on a slope having an angle of elevation of 25.0° . If pushing downhill on the block with a force just exceeding 187 N and parallel to the slope is sufficient to cause the block to start moving, find the coefficient of static friction.

Ans.: 0.850

29. A dockworker loading crates on a ship finds that a 20.0-kg crate, initially at rest on a horizontal surface, requires a 75.0-N horizontal force to set it in motion. However, after the crate is in motion, a horizontal force of 60.0 N is required to keep it moving with a constant speed. Find the coefficients of static and kinetic friction between crate and floor.

Ans.: $\mu_s = 0.383$, $\mu_k = 0.306$

31. **v** The coefficient of static friction between the 3.00-kg crate and the 35.0° incline of [Figure P4.31](#) is 0.300 . What minimum force $\vec{F}$ must be applied to the crate perpendicular to the incline to prevent the crate from sliding down the incline?



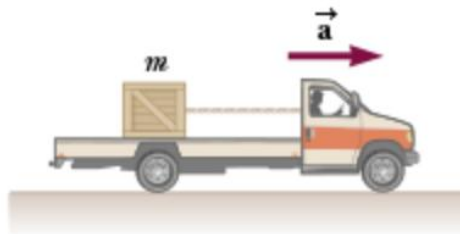
Ans.: 32.1 N

4.6 Tension Forces

33. A 75-kg man standing on a scale in an elevator notes that as the elevator rises, the scale reads 825 N . What is the acceleration of the elevator?

Ans.: 1.2 m/s^2 upward

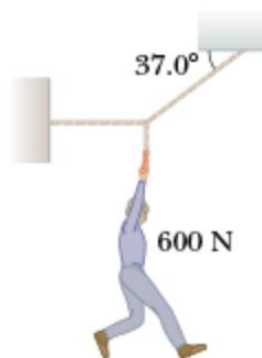
34. A crate of mass $m = 32 \text{ kg}$ rides on the bed of a truck attached by a cord to the back of the cab as in [Figure P4.34](#). The cord can withstand a maximum tension of 68 N before breaking. Neglecting friction between the crate and truck bed, find the maximum acceleration the truck can have before the cord breaks.



Ans.: $a_{\max} = 2.1 \text{ m/s}^2$

35. **Q/C**

a. Find the tension in each cable supporting the $6.00 \times 10^2 \text{ N}$ cat burglar in [Figure P4.35](#).



Ans.: $6.00 \times 10^2 \text{ N}$ in vertical cable, 997 N in inclined cable, 796 N in horizontal cable

36. The distance between two telephone poles is 50.0 m . When a 1.00-kg bird lands on the telephone wire midway between the poles, the wire sags 0.200 m . Draw a free-body diagram of the bird. How much tension does the bird produce in the wire? Ignore the weight of the wire.

Ans.: 613 N

37. **QIC**

- An elevator of mass m moving upward has two forces acting on it: the upward force of tension in the cable and the downward force due to gravity. When the elevator is accelerating upward, which is greater, T or w ?
- When the elevator is moving at a constant velocity upward, which is greater, T or w ?
- When the elevator is moving upward, but the acceleration is downward, which is greater, T or w ?
- Let the elevator have a mass of $1\,500\text{ kg}$ and an upward acceleration of 2.5 m/s^2 . Find T . Is your answer consistent with the answer to part (a)?
- The elevator of part (d) now moves with a constant upward velocity of 10 m/s . Find T . Is your answer consistent with your answer to part (b)?
- Having initially moved upward with a constant velocity, the elevator begins to accelerate downward at 1.50 m/s^2 . Find T . Is your answer consistent with your answer to part (c)?

Ans.: a. $T > w$ b. $T = w$ c. $T < w$
 d. $1.85 \times 10^4\text{ N}$; yes e. $1.47 \times 10^4\text{ N}$; yes f. $1.25 \times 10^4\text{ N}$; yes

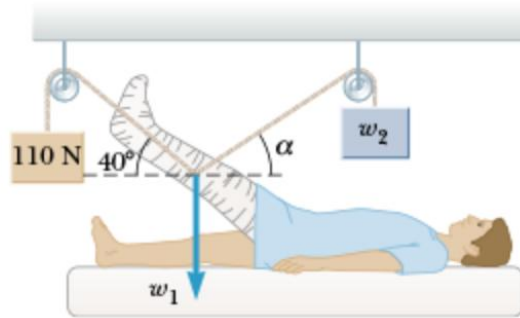
4.7 Applications of Newton's Laws

39. A 150-N bird feeder is supported by three cables as shown in [Figure P4.39](#). Find the tension in each cable.



Ans.: 150 N in vertical cable, 75 N in right-side cable, 130 N in left-side cable

40. **BIO** The leg and cast in [Figure P4.40](#) weigh 220 N (w_1). Determine the weight w_2 and the angle α needed so that no force is exerted on the hip joint by the leg plus the cast.



Ans.: $w_2 = 1.7 \times 10^2 \text{ N}$, $\alpha = 61^\circ$

43. Consider a commercial lorry carrying a heavy load, such as steel beams. A significant hazard for the driver is that the load may slide forward, crushing the cab, if the lorry stops suddenly in an accident or even in braking. Assume, for example, a $10\,000\text{-kg}$ load sits on the flatbed of a $20\,000\text{-kg}$ lorry moving at 12.0 m/s . Assume the load is not tied down to the lorry and has a coefficient of static friction of 0.500 with the flat bed.

- Calculate the minimum stopping distance for which the load will not slide forward relative to the lorry.
- Is any piece of data unnecessary for the solution?

Ans.: a. 14.7 m ; b. Neither mass is necessary.

44. A student decides to move a box of books into her dormitory room by pulling on a rope attached to the box. She pulls with a force of 80.0 N at an angle of 25.0° above the horizontal. The box has a mass of 25.0 kg , and the coefficient of kinetic friction between box and floor is 0.300 .

- Find the acceleration of the box.
- The student now starts moving the box up a 10.0° incline, keeping her 80.0 N force directed at 25.0° above the line of the incline. If the coefficient of friction is unchanged, what is the new acceleration of the box?

Ans.: a. 0.368 m/s^2 ; b. 1.28 m/s^2 down the incline

45. **Q1C** An object falling under the pull of gravity is acted upon by a frictional force of air resistance. The magnitude of this force is approximately proportional to the speed of the object, which can be written as $f = bv$. Assume $b = 15 \text{ kg/s}$ and $m = 50 \text{ kg}$.

a. What is the terminal speed the object reaches while falling?

b. Does your answer to part (a) depend on the initial speed of the object? Explain.

Ans.: a. 33 m/s ; \

b. No. The object will speed up to 33 m/s from any lower speed and will slow down to 33 m/s from any higher speed.

50. A car is traveling at 50.0 km/h on a flat highway.

a. If the coefficient of friction between road and tires on a rainy day is 0.100 , what is the minimum distance in which the car will stop?

b. What is the stopping distance when the surface is dry and the coefficient of friction is 0.600 ?

Ans.: a. 98.6 m ; b. 16.4 m

51. **V** A 5.0-kg bucket of water is raised from a well by a rope. If the upward acceleration of the bucket is 3.0 m/s^2 , find the force exerted by the rope on the bucket.

Ans.: 64 N